

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

DEPARTMENT OF CIVIL ENGINEERING

End Semester Examination (Monsoon Semester) (2025-26)

NCES101- Engineering Graphics

SET - 1

Date : 25-11-2025

Time : 09:00 AM to 11:00 AM

Duration : 2 hours

Total Marks : 40

Instructions

- There are 4 questions in this paper and each carry 10 marks. The questions are printed on both sides of the page.
- Use your space on drawing sheet efficiently & do not scale up/down the measurements. **However, if you need to do scale up/down, mention them clearly with scale factor.**
- All dimensions in the drawings/questions are in mm.
- Any missing dimension in the drawing must be assumed suitably.
- Layout dimension: 420 mm x 297 mm. Border size: 10 mm.**
- Draw the title block as provided below:

	120
10	NAME:
10	ADMN. NO:
10	SECTION:
10	SCALE:
40	

$$a \left(\frac{3}{2} - \frac{2}{3} \right) = 30$$

$$a \frac{5}{6} = 30 \quad a = 36$$

- The focus of a conic is 30 mm from directrix. Draw the locus of a point P moving in such a way that eccentricity is $\frac{2}{3}$. Draw a tangent and normal at a point on the conic at a distance of 10 mm along the major axis from the focus and lying in the upper half of the conic. [10]

- A cube of 35 mm long edges is so placed on HP on one of its corners such that the body diagonal is parallel to H.P. and perpendicular to V.P. Draw its projections. [10]

